VIGNESH KRISHNAKUMAR

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A highly motivated Data Analytics professional with a Master's in Data Analytics and a strong background in cybersecurity support and system optimization. Experienced in leading service improvement initiatives and managing large-scale system migrations. Proficient in statistical techniques, algorithms, and data visualization to support decision-making. Adept in using analytical tools to address practical issues, excited to use my knowledge of drawing conclusions from intricate datasets to support significant, data-driven solutions in a fast-paced, results-driven setting.

Education:

MSc Data Analytics | Aston University Jan 2024 – Jan 2025

Anthropogenic Impact on Marine Biodiversity:

- Analysed the human-made impact on biodiversity by studying correlations between environmental factors like Sea Surface Salinity (SSS) and Sea Surface Temperature (SST) with locality, latitude, and longitude.
- Applied machine learning techniques, including XGBoost and Random Forest, to predict variations in SSS and SST, with XGBoost achieving the highest accuracy of 90% on unseen data.
- Aligned with data transformation, predictive modelling, and optimization by leveraging advanced analytics to enhance data quality and performance in enterprise data environments.

Protein Sequence Analysis and Predictive Modelling for Epitope Detection:

- Built a predictive model for identifying B-cell epitopes in protein sequences, incorporating data cleaning (Pandas, NumPy), preprocessing, visualisation of data using python libraries such as Matplotlib, Seaborn and key feature selection using Fisher's score to enhance data quality and accuracy within structured datasets.
- Utilized SMOTE (Imblearn) for class balancing and applied Random Forest and XGBoost (Scikit-Learn) optimizing parameters for performance enhancement.
- Achieved an AUC of 0.56 on unseen data, demonstrating model generalization, scalability, and actionable insights while identifying areas for improvement in data-driven decision-making.

B.Tech Mechanical Engineering | SRM University Jun 2017 – Jun 2021

Captain and Vehicle Dynamics Head – Team Enigmatic Plodder:

- Optimized Quad Bike suspension for QBDC and Quad Torc, ensuring compliance. Led recruitment and training, fostering teamwork, problem-solving, and knowledge transfer for sustained team success.
- Managed cross-functional teams, financial oversight, manufacturing, and strategy. Led the team to consecutive QBDC championships, demonstrating leadership, collaboration, and resilience in high-pressure environments.

Mechanical and Metallurgical Characterisation of ARC Welded IN617 – SS304H:

- Optimized welding parameters for dissimilar metals (SS304H and IN617) using Double Pulse MIG welding, integrating structured data analysis and statistical modelling (Taguchi L9 design and ANOVA) to enhance process efficiency and quality, representing strong problem-solving and analytical skills.
- Applied multi-criteria decision-making (TOPSIS) to refine key welding factors, ensuring data-driven process optimization, aligning with ETL methodologies for structured data transformation and analysis, while fostering teamwork and collaboration in a research-driven environment.
- Achieved optimal welding conditions, supporting high-performance applications with reliable and validated insights

Skills:

Natural Language Processing (NLP): NLTK | Tokenization | Lemmatization & Stemming | Named Entity Recognition (NER) | Text Vectorization (Bag of Words, TF-IDF, Word2Vec) | Sentimental Analysis, Text Classification & User Behaviour Analysis (Project In-Progress on YouTube Comments Dataset from Kaggle)

Statistical & Machine Learning Algorithms: Linear & Logistic Regression | Bayesian Inference | Support Vector Machine | K – Nearest Neighbours
Clustering & Dimensionality Reduction: K-means Clustering | Principal Component Analysis
Anomaly Detection: Isolation Forest | One-Class SVM
Microsoft: Excel – (Learning Advanced excel techniques) | PowerBI – (Learning DAX Commands) | Azure Cloud.

Experience:

TATA Consultancy Services Nov 2021 – Dec 2023
Assistant Systems Engineer

- Designed and presented real-time monitoring dashboards, providing actionable insights to clients and improving client decision-making processes by 20%.
- Worked in an Agile environment, participating in daily stand-ups, sprint planning, and updates on service tickets, tasks to handover to colleagues.
- Collaborated with teams to analyse operational trends and implemented tailored solutions, enhancing service efficiency and exceeding client expectations.
- Provided data-backed recommendations for vulnerability management in servers and applications, improving overall system security and efficiency.
- Published knowledge-based articles on new error tickets and updated Standard Operating Procedures based to streamline operations and improve organizational learning.
- Led the end-to-end migration of legacy PKI servers to Windows Server 2019 collaborating with multiple technical teams, documenting system processes and test cases while ensuring seamless implementation, and adherence to SLAs. This also, included migrating Active Directory Certificate Services (ADCS), upgrading certificate templates, implementing secure communication protocols (SSL/TLS), and ensuring system stability post-migration.

TATA Consultancy Services Jun 2021 – Dec 2023
Security Deployment Lead

- Monitored and enforced organisational compliance policies, conducting regular workflow analyses to identify areas for improvement and ensure alignment with GDPR compliance standards.
- Managed onboarding and offboarding processes by leveraging Pivot Tables and VLOOKUP to efficiently organise and validate user data, ensuring accurate data handling and compliance with security policies while coordinating with cross-functional teams to meet organisational standards.
- Conducted and tracked compliance checks for 300+ machines, generating reports on risk analysis and providing actionable insights to rectify compliance gaps.
- Oversaw role-based access management, analysing access patterns to enhance data integrity and align processes with best practices for data governance and security.
- Led training sessions on data security and compliance protocols, promoting best practices and fostering a culture of ethical data use within the organisation.

Certification:

NVIDIA Fundamentals of Deep Learning (2025) (In-Progress)

Microsoft Azure Fundamentals (2022)

Hobbies:

Bouldering and Rock Climbing: A recently discovered activity which has endured my problem-solving skills and helped reduce screentime putting my physical and mental health in good space. It challenges my focus, builds strength, and offers a refreshing break that leaves me feeling recharged and balanced.

Online Gaming: An escape from reality during weekends for a couple of hours with friends in a fast-paced competitive environment involving strategic thinking.